EE3070 Project Proposal

Submission Instructions

1. Each group submits one copy **by your Team Leader** via Assignment in CANVAS.
2. File format:
   - Report in pdf; Filename: PPR-Lxx-GPy.pdf
     - Lxx: lab session number of Team Leader (L01/LA1, L02/LA2, L03/LA3)
     - GPy: group number
3. Deadline for submission:
   - For Tuesday sessions (L01/LA1, L02/LA2): 11pm 6th Oct 2026 (wk6)
   - For Monday sessions (L03/LA3): 11pm 5th Oct 2026 (wk6)

No late submission will be accepted. Please stay ALERT on CANVAS for any changes however unlikely on updates on submission deadlines.

Group No: L01/LA1 Gp 6

Name of the Project (Design a name to best reflect the outcome of your project; Don't just say Smart IoT)

Smart Library: Arduino Mega based seat availability and noise monitoring system.

Member Name	Group Leader Please give tick √	Student ID	Electronic Signature
Lam Si Hong	√	59299990	林
KUSSAIYN Olzhas		57885712	Olzhas
MUKHAMETKHAN Nurdana		58321615	Nurdana
Chan Chung Yin		58712976	Joe

Section 1: Description of Targeted Scenario and Project Goals

[Plan Phase: Identify Problem and Goal]

- **Introduce the background.**
- **Describe the identified problem(s).**
- **Describe the ethical constraints in your design.**
- **Describe the impact of your solution in a broad, global and societal context.**
- **Give a list of the goals and how users can be benefited from your project.**

This project is based on Arduino Mega and develops a system that monitors real time seat availability and noise conditions of the study areas. Individual seat will obtain Infra-red sensors to monitor whether seat is free or occupied, while mmWave sensor will be used to verify the result of IR sensor and detect the occupant of the seat is stationary. A microphone-based sensor will be used to classify the sound activity as silent, quiet and noisy. Additionally, the system will output the a live seat map, available seats count, noise state on OLED display.

Problems addressed:

- **Students often enter a library or study zones without knowing if there are any seats available or specific desks are free.**
- **Noise level detection will allow student to choose study zone appropriate to their goals or/and noise tolerance.**
- **Study areas noise policies may be introduced. This will solve the problem of a loud and disturbing students in quite study areas. This project will facilitate for detection and noise regulation of such areas.**

Ethical constraints: This project avoids using any computer vision implementation to preserve privacy protection; and rather uses infra-red and mmWave sensors.

This project aims to improve efficiency of university campus facilities and resource allocation, potentially reducing the time wasted on the search of the available seat. Furthermore, it promotes self-regulation of noise discipline and respect for other students in the study zones.

Goals:

- **Live seat availability tracking and visual seat map of the study zones.**
- **Noise monitoring system**
- **Configure adjustable noise policies: silent, quiet, collaborative study zones.**

Students benefit by saving time when finding available study space and being able to study in appropriate noise level environment.

Section 2: Description of System Functions

- **Describe the system functions that you are going to include in your project.**
- **For each function, state clearly what modules are used, and how modules are to be interacted to achieve the functions (better to give some block diagrams), how users use the functions**

Seat Availability Monitoring and Dashboard Query:
This function detects whether each monitored seat is available or occupied. Four GP2Y0A21YK0F infrared distance sensors are connected to the Arduino Mega 2560, with one sensor assigned to each seat. Each sensor intended for individual seat-level occupancy sensing.
The Arduino reads the analogue sensor signals, compares them with calibrated threshold values, and calculates the number of available seats. The processed data are sent through a Wi-Fi communication module to the web dashboard. User can access the dashboard by mobile phone or computer to check the number of free seats and the status of each monitored seat.

Noise Level Monitoring and Automatic Alert:
This function monitors the relative noise level in the study area. A MAX9814 microphone module captures ambient sound and sends an analogue signal to the Arduino Mega.
The Arduino processes the sound signal and classifies the environment as quiet, moderate, or noisy. This is a automatic function, users do not need to operate this function manually. When sustained excessive noise is detected, the system will issue a warning , ie. turns on a red LED/OLED warning or activates an optional short buzzer alert. To avoid false alarms from short sounds, such as chair movement or cough, an alert is activated only when high noise continues for a predefined period. The MAX9814 should be used to indicate relative sound activity, not an officially calibrated decibel value.

Section 3: Project Planning Schedule

Week	Tasks to be completed
6 …	Physical Model Construction & Seat Sensor Wiring: • Construct a four-seat study zone model representing a table at using cardboard or foam board. • Wire the four IR obstacle sensors under/around the desks.
7 …	Software implementation & Local Seat LEDs • Implement software like debounce logic to prevent false triggers from movements. • Wire and test the four individual seat status LEDs.
8 …	OLED Dashboard & Seat Map Software • Connect the OLED display. • Write the code to calculate free-seat counts and display the live seat map on the screen.
9 …	Audio Sensing, Calibration & Zone Modes • Integrate the analog microphone module. • Capture baseline silence, normal conversation, and loud activity to calibrate initial sound thresholds. • Implement push-button zone-mode switching for Silent, Quiet, and Collaborative policies.
10 …	Duration-Based Noise Filtering & Warning Feedback • Implement filtering for the microphone readings. • Program noise filtering to ignore brief sounds and trigger warnings only for continuous noise. • Integrate the RGB LED (Green/Yellow/Red) and the brief one-shot buzzer alert.
11 …	mmWave Radar Integration & Sensor Fusion • Wire the mmWave radar module using hardware UART. • Configure distance gates and test stationary human presence detection. • Program sensor-fusion rules to cross-check individual seat maps against zone-level radar presence and highlight any differences in results.
12 …	Testing & Preparation for the Presentation • Run tests measuring seat detection accuracy, false-positive rates, response times, and noise-alert timing errors. • Finalize wire management, physical labeling, and presentation/demonstration materials for the final project evaluation.
13 Final Presentation	

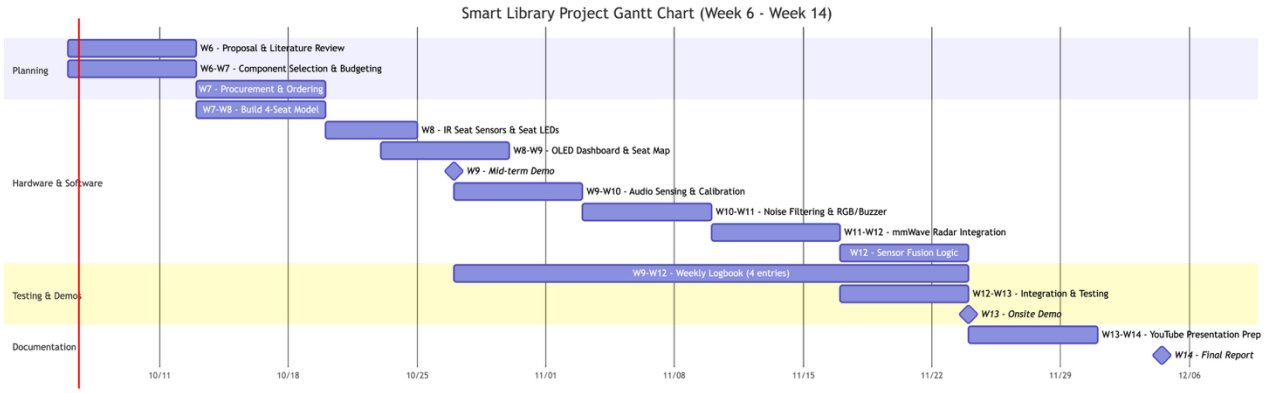
Section 4: Division of Labor (Explain clearly the responsibility of each member; which part will be designed by whom; etc.)

Member 1 Lam Si Hong 59299990	Design and build the physical 4-seat study zone model Mount, wire, and calibrate the IR/FSR seat sensors and individual desk LEDs Handle overall physical enclosure, labeling, and wiring neatness

Member 2 KUSSAIYN Olzhas 57885712	Program the Arduino Mega main controller loop and pin allocations Implement seat debounce logic and free-seat calculation algorithms Code and format the I2C OLED dashboard displays and seat maps
Member 3 MUKHAMETKHAN Nurdana 58321615	Configure and wire the LD2410 mmWave radar module via hardware UART and level shifters Tune range sensitivity gates for stationary student detection Develop the sensor-fusion logic to detect and display disagreements between seat maps and zone presence
Member 4 Chan Chung Yin 58712976	Integrate the analog microphone module and calibrate ambient sound baselines Implement moving-average amplitude calculations and duration-based noise filtering Set up the RGB LED and non-disruptive buzzer warning system. Design and execute the quantitative test plan (accuracy, false positives, response time) and compile results

Section 5: Gantt Chart:

Provide a Gantt Chart to illustrate schedule and participation of members, as given in Sec. 3 and 4.



Section 6: Summarize the current progress (eg. what each member has tried, prepared or done)

Member 1 Lam Si Hong 59299990	Ordering sensors and modules for the project. Project workflow, labor allocation and Gantt Chart.
Member 2 KUSSAIYN Olzhas 57885712	Research of the specific sensors needed. Identifying suitable project scope, hardware choice and system design
Member 3 MUKHAMETKHAN Nurdana 58321615	Background and problem research.
Member 4 Chan Chung Yin 58712976	Research of the specific sensors needed. Planning of system functions.

Marking Rubric:

Criteria	A (Excellent)	B (Good)	C (Satisfactory)	D (Poor)	F (Fail)
Problem Identification	Significant problem, clear in description, well thought on conditions, objectives/scope well defined	Clear in description, objectives/scope properly defined	Clear in some items and requiring further improvement	Unclear in most, needs further improvement	Unclear and inappropriate
Ethical constraints	Responsibility to the profession, colleagues, employers or clients and the public have been considered.	Responsivities to the major parties involved have been considered.	Responsibilities to some parties have been considered.	Responsibilities to minor parties have been considered.	No consideration to any parties have been considered.
Problem Solving	Suitable methods are proposed and the problem would be fully solved	Suitable methods are proposed and the problem would reasonably be solved. Some technical techniques are noted.	Methods are proposed but they would only partially solve the problem.	Methods are proposed but they can't solve the problem in an appropriate way.	No method was suggested to solve the problem
Solution Impact	Solution has a major impact in a broad, global and societal context.	Solution has a significant impact in a broad, global and societal context.	Solution has a major impact either broadly, globally of societal in context	Solution has a minor impact, either narrow in scope and of local significance.	Solution has no impact.
Task Ident & Project management	Tasks are clearly described and appropriately identified. Good use of project management and tools. Division of labor is good and well-balance	Tasks are identified and described in a reasonable way. Some use of project management and tools, but can be improved. Division of labor is reasonable.	Tasks are identified, but need to be improved. Specified tasks are not well described. Apply project management in a limited way. Division of labor is not balanced.	Tasks are vague and inappropriate. Little evidence of project management. Division of labor has obvious problems.	No schedule and plan
Writing & Grammar	Spell-checked and proofread well. Writing style indicates planning that makes reading easy and flow of material makes understanding easy.	A few spelling and grammatical errors. Writing style indicates planning that makes reading easy.	More than one spelling/ grammar error per section. Readable writing style, but difficult to follow.	Frequent spelling and grammatical errors. Writing disorganized and difficult to read and understand.	No attempt